

Introduction to Radiobiological Models



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- 1 Linear Quadratic (LQ) Model
- 2 Nominal Standard Dose (NSD)
- 3 Cumulative Radiation Effect (CRE)
- 4 Time-Dose-Fractionation (TDF)

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- 1 Linear Quadratic (LQ) Model
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- Post-irradiation **cell survival** as a function of radiation dose can be **biophysically modeled** in several different ways.
- **Poisson statistics** are used for the basis for survival equations.

Mathematical Modeling

Poisson Statistics

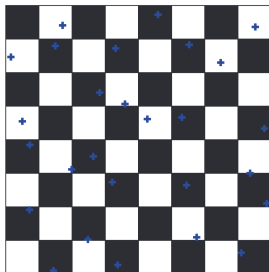


Figure: If it rains on a chessboard, and an average of x drops hits each square, How many squares are dry?

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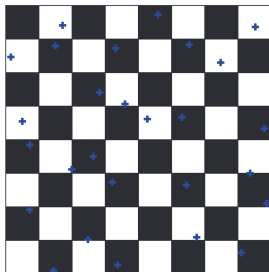


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- Poisson statistics describe a large number of random events happening to a large number of subjects, averaging out to a small number of events per subject.

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$$P(k) = \frac{x^k e^{-x}}{k!}$$

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- The probability of a square being wet is given by:

$$P(\text{wet}) = 1 - e^{-x}$$

Poisson Distribution

Cell Survival

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 - 1 hit kills 63% of cells. $SF = 0.37$

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Note

D_0 is defined as the radiation dose resulting in 37% survival. This is assumed to be equal to the radiation dose necessary to cause one **lethal** hit per cell.

Poisson Distribution

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Note

To achieve a certain TCP, you should aim for a tumor cell survival of $(1 - TCP)$.

Target theory

Single-Target, Single-Hit Model

- It assumes that each cell has a **single critical target**, and that a single hit to this target is sufficient to cause cell death.

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where D is the dose and D_0 is the dose required to reduce the surviving fraction to 37%.

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- This type of survival curve is seen for mammalian cells which are irradiated with **high LET** radiation such as alpha particles or carbon ions; very **radiation-sensitive** cells like lymphocytes and bone marrow cells; cells that have major defects in DNA doublestrand break repair; are synchronized in M phase; and for cells that have DNA double-strand break repair inhibited by chemical or gene knockdown.

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- For a single dose D , the survival fraction (SF) is given by:

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where D is the dose, D_0 is the dose required to reduce the surviving fraction to 37%, and n is the number of targets.

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- This type of survival curve is seen for mammalian cells which are irradiated with **low LET** radiation such as X-rays or gamma rays; cells that are **radiation-resistant** like fibroblasts and epithelial cells; cells that have a normal DNA double-strand break repair; and for cells that are synchronized in G1 or S phase.

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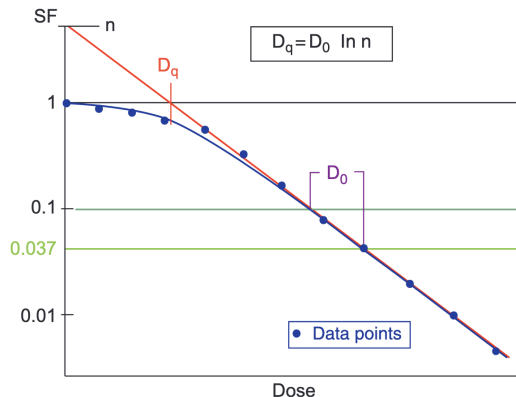


Figure: The single-hit, multitarget survival curve. Notice that it is curved at low doses and straight (e.g., survival is an exponential function of dose) at high doses. This allows you to calculate D_0 by taking measurements in the high-dose region

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Important to note

Do not measure D_0 at $SF = 0.37$ if the survival curve has a shoulder. Remember that D_0 applies to the high-dose portion of the curve in that case! Instead, measure the difference in dose between $SF = 0.1$ and $SF = 0.037$.

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- $D_{10} = 2.3 \times D_0 =$ a dose that will reduce surviving fraction by tenfold (i.e., from $SF = 0.1$ to $SF = 0.01$).

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- The low-dose portion of the curve is known as the “shoulder”:
 - D_q tells you how wide the shoulder is.
 - D_q is a measure of the cell's repair capacity. More repair means a larger D_q and a larger shoulder

Single-Hit, Multitarget Model

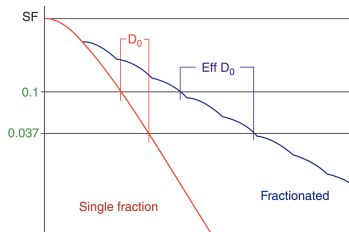
Advantages and Disadvantages

Fractionated Radiation and Effective D_0

- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.

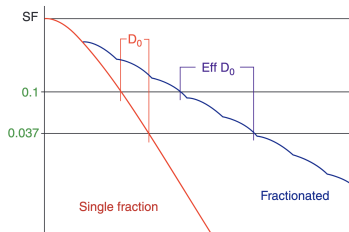
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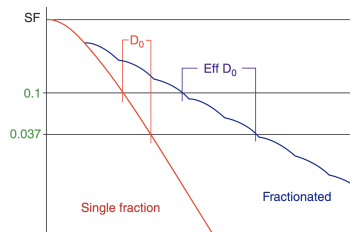
- A fractionated radiation survival curve has a shallower slope compared to a single-fraction survival curve.
- Assuming complete repair in between fractions, the shoulder is repeated with each fraction.
- D_0 versus effective D_0 . When radiation is fractionated, you need much more dose to achieve the same amount of cell kill. Therefore, effective D_0 is always larger than D_0



Fractionated Radiation and Effective D_0

Continue

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Fractionated Radiation and Effective D_0

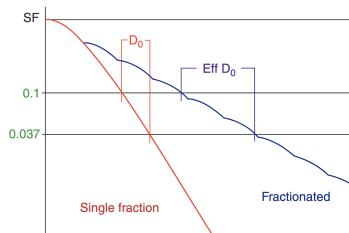
Continue

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- For a single fraction of D Gy with a surviving fraction of SF_D :

$$SF_D = e^{-D/D_0}$$

$$SF_D = e^{-D/\text{effective } D_0}$$

$$\text{effective } D_0 = -\frac{D}{\ln(SF_D)}$$



Fractionated Radiation and Effective D_0

Continue

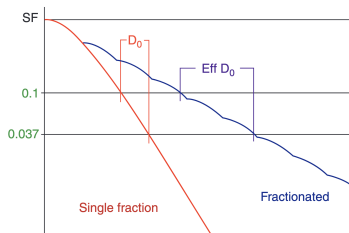
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- After x fractions to total dose xD Gy:
 $SF_x^D = e^{-xD/\text{effective } D_0}$



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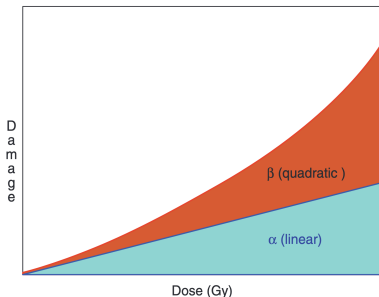
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 - Single-hit kill (α) is unrepairable damage and is **independent** of fractionation or dose rate. This corresponds to single-hit and **intra**-track accumulated damage.
 - Multi-hit kill (β) is repairable damage and is **dependent** on fractionation or dose rate. This corresponds to double-hit and **inter**-track accumulated damage.

Linear-Quadratic (LQ, Alpha-Beta) Model

Continue:

- The α/β ratio is the **dose** at which α kill and β kill are equal.
 - Low α/β ratio (“high repair”) tissues are relatively resistant at small fraction size and relatively sensitive at large fraction size.
 - High α/β ratio (“low repair”) tissues are relatively sensitive at small fraction size and relatively resistant at large fraction size.



Linear-Quadratic (LQ, Alpha-Beta) Model

Alpha-Beta Model and Dose Fractionation

- LQ model is widely used in radiotherapy to predict the effects of different fractionation schemes on both tumor and normal tissues.
- Suppose we have n no of fractions of dose d each, then the total dose $D = nd$. The surviving fraction after n fractions is given by:

$$\begin{aligned} SF_n &= e^{-n(\alpha d + \beta d^2)} \\ \Rightarrow -\frac{1}{nd} &= \frac{\alpha}{\ln(SF_n)} + \frac{\beta d}{\ln(SF_n)} \end{aligned}$$

- Plot of dose per fraction d versus $-\frac{1}{nd}$ gives a straight line with slope $\beta/\ln(SF_n)$ and y-intercept $\ln(SF_n)$. The ratio of intercept to slope gives α/β .
- Such values are obtained for a variety of tissues and tumors, and are used to guide clinical decision-making regarding fractionation schemes.

Linear-Quadratic (LQ, Alpha-Beta) Model

α/β Ratios of Tissues and Tumor

- Most acute-reacting tissues are believed to have an α/β ratio ≈ 10 .
- Most late-reacting tissues are believed to have an α/β ratio ≈ 3 or less.
- CNS tissue (brain, cord) is believed to have an α/β ratio $\approx 1-2$.
- Most tumors are believed to have an α/β ratio ≥ 10 .
- Some low α/β tumors may have an α/β ratio $\approx 1.5-4$. Breast and Prostate are the “classic” low α/β tumors.

Linear-Quadratic (LQ, Alpha-Beta) Model

Biologically Effective Dose

- One advantage of the linear-quadratic model is that it is easy to calculate effective doses for different fraction sizes.

Definition

Biologically effective dose (BED) is an extrapolated dose given over infinitely many fractions.

- For n fractions of d dose per fraction:

$$\begin{aligned}\text{BED}_{\alpha/\beta} &= nd \left(1 + \frac{d}{\alpha/\beta} \right) \\ &= D \left(1 + \frac{d}{\alpha/\beta} \right)\end{aligned}$$

- $\text{BED} = \text{total physical dose} \times \text{relative effectiveness of that dose.}$

Linear-Quadratic (LQ, Alpha-Beta) Model

Practical Applications

Problem 1

If a treatment is planned for 70 Gy/ 35 fractions, but due to departmental constraints, we need to complete it in 25 fractions. What is the new dose per fraction for same tumor control with $\alpha/\beta = 10$?

Linear-Quadratic (LQ, Alpha-Beta) Model

Equivalent Dose in 2 Gy fractions (EQD₂)

- EQD₂ is the dose in 2 Gy fractions that would produce the same biological effect as the actual treatment.
- For n fractions of d dose per fraction:

$$\text{EQD}_2 = \frac{BED}{1 + \frac{2}{\alpha/\beta}}$$
$$\Rightarrow \text{EQD}_2 = D \left(\frac{d + \alpha/\beta}{2 + \alpha/\beta} \right)$$

Linear-Quadratic (LQ, Alpha-Beta) Model

Practical Applications

Problem 2

You want to treat a lung cancer to 60 Gy in 2 Gy fractions, but the first five fractions are given at 3 Gy/fx due to SVC syndrome. Assuming $\alpha/\beta = 3$, how much additional dose should you deliver at 2 Gy per fraction?

Question 1: Which of the following remains unaltered while changing the plan from fractionated radiotherapy to brachytherapy?

- ① Total Dose
- ② Dose per fraction
- ③ BED
- ④ Relative effectiveness of that dose

Question 1: Which of the following remains unaltered while changing the plan from fractionated radiotherapy to brachytherapy?

3 BED

MCQ Example

Question 2: A tumour containing 10^8 cells is inactivated by radiation with a mean lethal dose of $D_0 = 1.8$ Gy. What is the number of surviving tumour cells after a radiation dose of 45 Gy?

- ① 1.2×10^{-3}
- ② 1.4×10^{-4}
- ③ 1.4×10^{-3}
- ④ 1.2×10^{-4}

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Solution

The number of surviving cells is given by:

$$N = N_0 e^{-D/D_0}$$

where N_0 is the initial number of cells, D is the dose, and D_0 is the mean lethal dose. Substituting the values, we get:

$$N = 10^8 e^{-45/1.8} \approx 1.4 \times 10^{-3}$$

Question 3: It is required to replace an LDR implant of 60 Gy at 0.37 Gy/hr by a 3 fraction HDR implant. What dose/fraction should be used to keep the effect on the tumour the same, assuming $\alpha/\beta = 15$ and $\mu = 1.4 \text{ hr}^{-1}$?

- ① 10.25 Gy
- ② 11.32 Gy
- ③ 11.65
- ④ 10.92

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